

# Spring 2025 Practice Midterm Exam

## Foundations of Data Science

Name

## Instructions

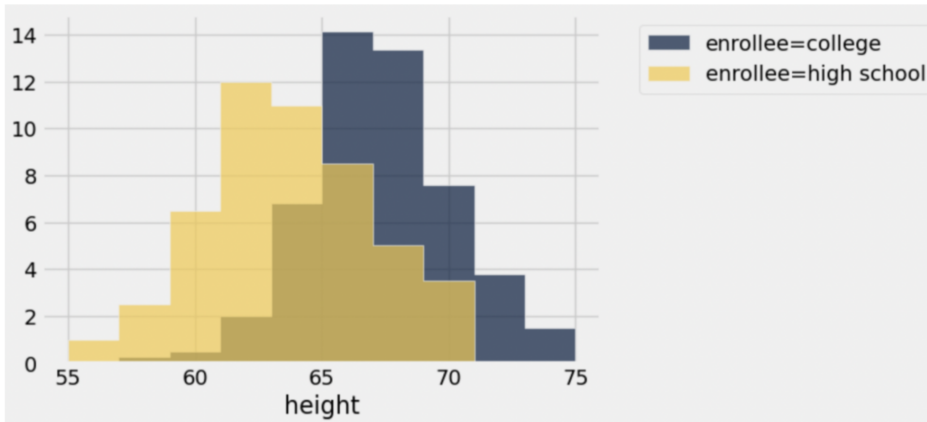
- Open the Midterm Exam Reference Guide (linked in Canvas). We will provide you with a new copy of this reference to use on the exam.
- Previews of most of the tables you will work with in this exam are at the end of exam in the Table Reference section.
- You can assume the following code has been run, when you are writing your Python code:

```
from datascience import *
import numpy as np
import matplotlib+
%matplotlib inline
import matplotlib.pyplot as plt
plt.style.use('fivethirtyeight')
```

- Select the correct response(s) or provide a written response depending on the question type. If a prompt implies you should write code, then you can provide your own code or use the provided template. Try to provide your responses in the template blanks or boxed spaces provided. If you find that you need additional space, write your extended response(s) on a blank sheet of paper and number them, so we can connect your response to the question.
- Once you have made an attempt on the exam, upload a PDF of your attempt to Canvas for a Complete score. Sample solutions will be released before the Midterm Exam Study Session.
- You will be asked a few questions about Project 1. Some of the questions may be about the code you wrote, and some of the questions may have you interpret the results and visualizations.

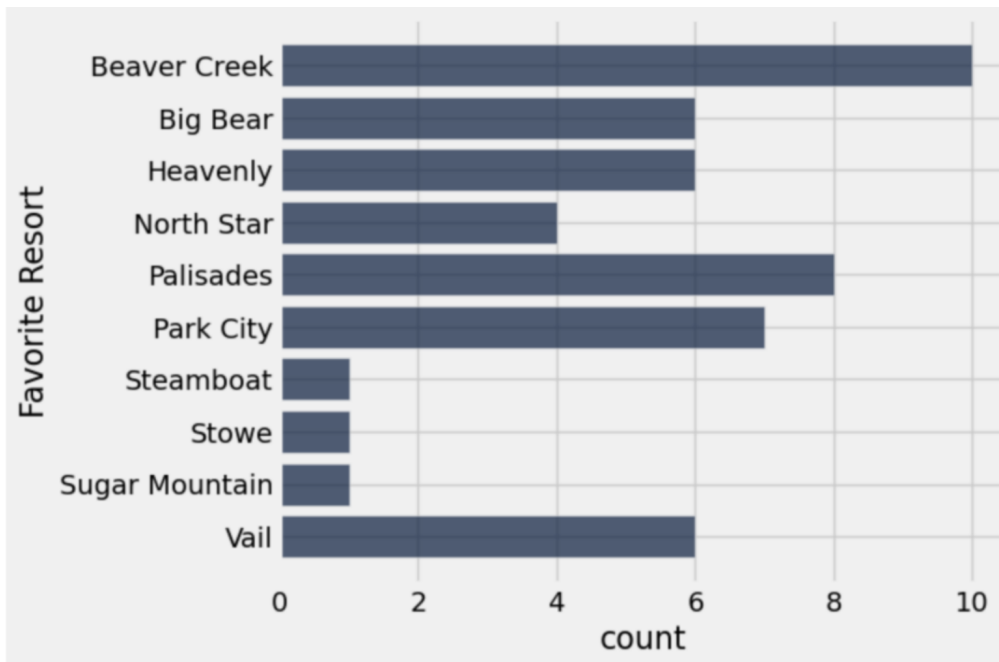
This practice exam contains questions from past UC Berkeley DATA 8 midterm exams in addition to a few additional or modified questions.

1. Dylan, a Data 8 student, wonders if there is a difference in heights between his high school and college classmates. Among Data 8 students, he collects a random sample of 100 high school students and a random sample of 200 college students, and organizes his data in a table called `heights`. He created the following histogram based on the data in `heights` where the  $x$ -axis of the histogram has a range of  $[55, 75]$  and each bin is 2 inches in width.



- (a) Unfortunately, Dylan forgot to include the y-axis label in his histogram. Which of the following is an appropriate y-axis label? Select one.
- ☐ Percentage of students
  - ☐ Percentage of students per height
  - ☒ **Percentage of students per inch**
  - ☐ Rate of change in height
- (b) Which of the following is closest to the total number of individuals in our table that are greater than or equal to 61 inches, but less than 63 inches in height? Select one.
- ☐ 14
  - ☐ 24
  - ☐ 28
  - ☒ **32**
- (c) Which of the following conclusions can you draw from Dylan's overlaid histograms? Select all that apply.
- ☐ Assuming we know the height of each bin, we can calculate the percentage of individuals in any range of heights.
  - ☒ **College Data 8 students are generally taller than high school Data 8 students.**
  - ☐ The area of both histograms together sum to 100%, so the area of the blue and yellow histograms are  $\frac{100}{300} \times 100$  and  $\frac{200}{300} \times 100$ , respectively.
  - ☐ There is an association between a Berkeley student's height and age.
  - ☐ College students enrolled in Data 8 are generally taller than high school students in Data 8 because they are older.
2. Kanu has been hired as a data scientist for Cal Rec Sports! His primary task is to analyze aquatic equipment rental data for the Berkeley Marina. Kanu's team presents him with a `rentals` table that contains the daily number of kayak and windsurfing board rentals, along with information pertaining to the day that the data were collected.

- (a) Which single visualization is most appropriate to describe the distribution of wind speeds in **rentals**. Select one.
- ☒ **Histogram**
  - ☐ Overlaid Histograms
  - ☐ Scatter Plot
  - ☐ Overlaid Scatter Plots
  - ☐ Bar chart
  - ☐ Line Plot
- (b) How does the distribution of kayak rentals on weekdays compare to the distribution of kayak rentals on weekends? Choose which single visualization is most useful for answering this question. Select one.
- ☐ Histogram
  - ☒ **Overlaid Histograms**
  - ☐ Scatter Plot
  - ☐ Overlaid Scatter Plots
  - ☐ Bar chart
  - ☐ Line Plot
3. Suppose we have discovered an association between two variables in a dataset. Which of the following would be the best way to test whether it is causal? Choose one.
- ☐ Use hypothesis testing to check whether the association is statistically significant.
  - ☒ **Run a randomized controlled experiment.**
  - ☐ Brainstorm some potential confounding factors and test whether any of them has an association with both variables.
  - ☐ If both variables are numerical, use a scatter plot to check for a trend.
4. The table **skiers** previewed on the Table Reference page contains information about the preferences of several skiers from a convenience sample of Data 8 staff.
- (a) Fill in the blanks to generate the following bar chart showing the popularity of everyone's Favorite Resort given in the skiers table. You may find the axis labels helpful.



```
favorite_counts = skiers._____ (_____)
_____ ._____ (_____)
```

#### Sample Solution:

```
favorite_counts = skiers.group("Favorite Resort")
favorite_counts.barh("Favorite Resort")
```

- (b) The height of the skiers was accidentally inputted into the `skiers` table as strings. Create a function called `str_to_int` that takes an input of a skier height in the format of a string and outputs the height as an integer. For example `str_to_int('71')` will output 71.

#### Sample Solution:

```
def str_to_int(height):
    return int(height)
```

- (c) Update the table `skiers` so that the column 'Height (in)' has integer values, not string values.

Hint. Consider using the `apply` and `with_column` table methods.

**Sample Solution:**

```
heights = skiers.apply(str_to_int, 'Height (in)')
skiers = skiers.with_column('Height (in)', heights)
```

- (d) Compute the average skier height for all the skiers in the `skiers` table.

**Sample Solution:**

```
np.average(skiers.column('Height (in)'))
```

5. If `customers` is a Table with a column called `'Age'`, why does the following code produce an error?

```
customers.where('Age', are.above(24)) / customers.num_rows
```

**Sample Solution:** This produces an error because `customers.where('Age', are.above(24))` produces another Table and `customers.num_rows` produces an integer, yet the division of a Table and an integer is not defined.

6. The empirical distribution of a categorical variable contains the unique values of the variable along with how often each value is observed.

☒ **True**

☐ False

7. The table `orders` previewed on the Table Reference page contains information about food orders that members of Data 8 Course Staff have made this semester.

- (a) Create a scatter plot to visualize the relationship between how much the order costs versus how the user rated the order.

```
orders._____ (_____, _____)
```

**Sample Solution:**

```
orders.scatter("Total", "Rating")
```

- (b) Assign the variable `usually_friends` to `True` if ordering with friends is more common than not ordering with friends and `False` otherwise.

```
with_friends = orders._____ (_____, True).num_rows  
without_friends = _____ - _____  
usually_friends = with_friends _____ without_friends
```

**Sample Solution:**

```
with_friends = orders.where("With Friends", True).num_rows  
without_friends = orders.num_rows - with_friends  
usually_friends = with_friends > without_friends
```

- (c) Assign the variable `frugal_user` to the name of the user who has spent the least over the entire semester:

```
frugal_user = (  
    orders.group(_____, _____)  
    .sort(_____, descending=_____)  
    .column("User").item(0)  
)
```

### Sample Solution:

```
frugal_user = (  
    orders.group("User", sum)  
    .sort("Total sum", descending=False)  
    .column("User").item(0)  
)
```

8. For this problem we are considering restaurants around Berkeley. The **restaurant** table contains information about specific restaurants including their distance from campus in miles. There are no duplicate restaurants in this table. Additionally, the **transport** table contains information about how long it takes to get to each restaurant using various modes of transportation. Each restaurant may appear multiple times in this table with different modes of transportation and time in minutes.

(a) Which code snippet would produce a table containing the fastest time for any type of food (e.g., Pizza, Bagels, Boba, ...)?

- ☐ (transport  
    .select("Type", "Time")  
    .group("Type", min))
- ✓ (restaurant  
    .join("Restaurant", transport, "Restaurant")  
    .select("Type", "Time")  
    .group("Type", min))
- ☐ (restaurant  
    .join("Restaurant", transport, "Restaurant")  
    .pivot("Type", "Time"))

(b) Which code snippet would produce a table with columns corresponding to each unique transportation mode (e.g., "Bus", "Drive", ...), rows corresponding to each unique restaurant type (e.g., Pizza, Bagels, Boba, ...) and the cells containing the minimum travel time.

- ☐ (restaurant  
    .join("Restaurant", transport, "Restaurant")  
    .select("Transportation", "Type", "Time")  
    .group("Transportation", "Type", min))
- ☐ (restaurant  
    .pivot("Transportation", "Type", "Time", min)  
    .join("Restaurant", transport, "Restaurant"))
- ✓ (restaurant  
    .join("Restaurant", transport, "Restaurant")  
    .pivot("Transportation", "Type", "Time", min))

9. Which of the following will be output by running the following block of code?

```
x = 0
if x == 0:
    x = 1
if x == 1:
    x = 2
elif x < 3:
    x = 3
else:
    x = 0
print("x is", x)
```

- ☐ x is 0
- ☐ x is 1
- ☒ x is 2
- ☐ x is 3

10. Berkeley adds a photo of a staff members pet at the end of each lab assignment. Each pet photo is chosen from a collection of 20 pets with 10 cats, 9 dogs, and 1 bird. For each event below, choose the Python expression that evaluates to the probability of that event. Select one.

(a) When one pet is chosen at random, the probability that it is either a cat or a bird.

- ☐  $(9 / 20) ** 2$
- ☐  $(10 / 20) * (1 / 20)$
- ☒  $(10 / 20) + (1 / 20)$
- ☐  $1 - (9 / 20) ** 2$
- ☐  $1 - (10 / 20) * (1 / 20)$
- ☐  $1 - ((10 / 20) + (1 / 20))$

(b) When two pets are chosen at random with replacement, the probability that they are both dogs.

- ☒  $(9 / 20) ** 2$
- ☐  $(10 / 20) * (1 / 20)$
- ☐  $(10 / 20) + (1 / 20)$
- ☐  $1 - (9 / 20) ** 2$
- ☐  $1 - (10 / 20) * (1 / 20)$
- ☐  $1 - (10 / 20) + (1 / 20)$

(c) When two pets are chosen at random with replacement, the probability that the first is a cat and the second is not.

- ☐  $10 / 20 + 10 / 20$
- ☒  $(10 / 20) * (10 / 20)$
- ☐  $(10 / 20) * (9 / 20) * (1 / 20)$
- ☐  $1 - (10 / 20) * (10 / 20)$
- ☐  $1 - (10 / 20 + 10 / 20)$
- ☐  $1 - (10 / 20) * (9 / 20) * (1 / 20)$



- (d) When two pets are chosen at random with replacement, the probability that the first chases the second. Assume dogs only chase cats, cats only chase birds, and birds don't chase.

- ☐  $(19 / 20) * (10 / 20)$   
☒  $(10 / 20) * (1 / 20) + (9 / 20) * (10 / 20)$   
☐  $1 - ((9 / 20) * (1 / 20) + (10 / 20) * (9 / 20))$   
☐  $1 - ((10 / 20) ** 2 + (9 / 20) ** 2 + (1 / 20) ** 2)$   
☐  $1 - ((10 / 20) ** 2 + (9 / 20) ** 2 + (1 / 20))$

11. The pygmy hippo is a small, reclusive (and cute) hippopotamid type that is native to the forests and swamps of West Africa. Two teams of zoologists set out to estimate the proportion that are male by sampling at random from the population. The first team samples 100 hippos and finds the proportion of males in their sample to be A. The second team samples 40 hippos and finds the proportion of males in their sample to be B. The full population has all 2,500 wild pygmy hippos; the proportion P of males in the population is 50% (but unknown to the zoologists).

- (a) Which of the following are more likely than not? Select all that apply.

- ☐ A is smaller than B.  
☐ A is larger than B.  
☒ **P is closer to A than B.**  
☐ P is closer to B than A.  
☐ None of these.

- (b) Which of the following is largest?

- ☐ The chance that A is above 55%  
☒ **The chance that B is above 55%**  
☐ The chance that A is above 60%  
☐ The chance that B is above 60%

12. Complete the code below to simulate rolling a fair six-sided die 10,000 times and estimate the probability of rolling a 4 or higher.

```
count = 0
num_reps = 10_000
die = np.arange(1, 7)

for i in np.arange(_____):
    random_die_roll = np.random.choice(_____)
    if _____:
        count = count + 1

prob_4_or_higher = _____
```

### Sample Solution:

```
count = 0
num_reps = 100
die = np.arange(1, 7)

for i in np.arange(num_reps):
    random_die_roll = np.random.choice(die)
    if random_die_roll >= 4:
        count = count + 1

probab_4_or_higher = count / num_reps
```

13. In the game of Wordle, a player guesses up to 6 words until they correctly guess the secret word of the day or run out of guesses. Their guess count is either the guess number that was correct, 1 through 6, or X if all 6 guesses were incorrect.

The table called `wordle` contains three columns. The first column shows the possible number of guesses. The second column shows the proportion of guess counts for all the players who tried to guess yesterday's word. The third column shows the proportion of all 1,000 Berkeley students with each guess count for yesterday's word. Here is a preview of the table:

| Count | Everyone | Berkeley Students |
|-------|----------|-------------------|
| 1     | 0.0      | 0.0               |
| 2     | 0.17     | 0.09              |
| 3     | 0.33     | 0.25              |
| 4     | 0.27     | 0.32              |
| 5     | 0.20     | 0.28              |
| 6     | 0.02     | 0.03              |
| X     | 0.01     | 0.03              |

The following questions are part of the steps we would use to investigate whether the distribution of guess counts for UC Berkeley students differs from the distribution for all players on yesterday's Wordle.

- (a) Complete the null hypothesis: "The distribution of guess counts for UC Berkeley students is ..."
- ☐ uniform with a 1/7 chance for each possible guess count.
  - ☐ like a random sample from a uniform distribution with a 1/7 chance for each possible guess count.
  - ☐ different from a uniform distribution with a 1/7 chance for each possible guess count.
  - ☐ the population of guess counts for all Wordle players.
  - ☒ **like a random sample from the population of guess counts for all Wordle players.**
  - ☐ different from the population of guess counts for all Wordle players.

- (b) Complete the alternative hypothesis: "The distribution of guess counts for UC Berkeley students is ..."
- ☐ the same as the distribution of guess counts for all Wordle players.
  - ☒ **different from the distribution of guess counts for all Wordle players.**
  - ☐ the same as the uniform distribution.
  - ☐ different from the uniform distribution.
- (c) Which test statistic is best for choosing between the null and alternative hypotheses?
- ☐ total guess count
  - ☐ most common guess count
  - ☐ guess count
  - ☒ **total variation distance**
  - ☐ observed average
- (d) Which line of code simulates a distribution of proportions for 1000 Berkeley students under the null hypothesis?
- ☐ `sample_proportions(1000, wordle.column(2))`
  - ☒ `sample_proportions(1000, wordle.column(1))`
  - ☐ `sample_proportions(1000, make_array('1', '2', '3', '4', '5', '6', 'X'))`
  - ☐ `sample_proportions(1000, make_array(1/7, 1/7, 1/7, 1/7, 1/7, 1/7, 1/7))`
- (e) Assume the observed test statistic is assigned to `obs`. We simulate under the null hypothesis 10,000 times and append each simulated test statistics to an array `sim`. Complete the following expression that computes the p-value for this hypothesis test.
- \_\_\_\_\_ ( \_\_\_\_\_ >= \_\_\_\_\_ ) / len( \_\_\_\_\_ )

**Sample Solution:**

```
np.count_nonzero(sim >= obs) / len(sim)
```

- (f) If the null hypothesis is rejected because the p-value of this hypothesis test is very small, what can we conclude? Select all that apply.
- ☐ Attending Berkeley improves most people's Wordle performance.
  - ☐ Attending Berkeley changes most people's Wordle performance.
  - ☐ Attending Berkeley does not improve most people's Wordle performance.
  - ☐ Attending Berkeley does not change most people's Wordle performance.
  - ☒ **None of these.**
14. Suppose a hypothesis test is proposed and we already know that the null hypothesis is true. If 500 researchers each independently collect a sample of the same size to carry out an experiment and they all use 1% as their p-value cutoff, we should expect around 5 of them to reject the null.
- ☒ **True**
  - ☐ False

15. George Michael Bluth has graduated from high school and is now living his lifelong dream: working at his family's frozen banana stand in Newport Beach. He's noticed that the volume and makeup of orders during the school year is quite different than during the summer. His uncle, Buster Bluth (an archeological data scientist), decides to help analyze these differences in human behavior by sampling sales data throughout the year and putting this information into the **bananas** table.

George Michael believes that banana sales are lower during the school year. He cites that during the school year they sell, on average, 25 fewer bananas per day than during the summer. However, Buster believes that this difference is only due to chance. Use the following hypotheses for the next questions:

- Null Hypothesis: The average number of bananas sold per day at a banana stand in Newport Beach is the same during the school year as it is during the summer.
  - Alternative Hypothesis: The average number of bananas sold per day at a banana stand in Newport Beach is lower during the school year than it is during the summer.
- (a) George Michael decides to simulate 1,000 times under the null hypothesis and store the test statistics from each simulation in the array **test\_stats**. For his test statistic, he chooses the average number of bananas sold during the school year minus the average number of bananas sold during the summer. He writes the following partially completed code:

```
def one_simulation():
    shuffled_labels = _____
    data_with_shuffled = _____
    average_by_label = _____
    averages = _____
    return _____

num_simulations = 1000
test_stats = make_array()

for _____:
    _____
```

### Sample Solution:

```
def one_simulation():
    shuffled_labels = bananas.sample(with_replacement=False).column('Session')
    data_with_shuffled = bananas.with_column('Shuffled', shuffled_labels)
    average_by_label = data_with_shuffled.group('Shuffled', np.average)
    averages = average_by_label.column('Sales average')
    return averages.item(0) - averages.item(1)

num_simulations = 1000
test_stats = make_array()

for i in np.arange(1_000):
    test_stats = np.append(test_stats, one_simulation())
```

- (b) If he obtained a p-value of 0.013 and the p-value cutoff is 5%, which of the following can he conclude about the hypothesis test?
- ☐ The data are consistent with the null hypothesis.
  - ☒ **The data are consistent with the alternative hypothesis.**
  - ☐ There is not enough information to make a conclusion of any kind.

16. Do you have questions about Project 1?

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# Table Reference

## orders

Here is a preview of the table `orders`:

| User      | Restaurant  | Total | Receiver       | Rating | With Friends |
|-----------|-------------|-------|----------------|--------|--------------|
| w3ndyk1m  | Sharetea    | 5.40  | stephaniekeem  | 10     | True         |
| s_kw33    | Riceful     | 10.24 | haileyebonjung | 2      | False        |
| nikkyp    | La Burrita  | 14.98 | wfurtaco       | 7      | True         |
| sonyaki55 | Poke Parlor | 12.86 | oskibear       | 8      | False        |

... (46 rows omitted)

The table has 6 columns:

- **User:** (string) username of the user who purchased the order
- **Restaurant:** (string) restaurant name of the order
- **Total:** (float) total amount spent on the order, in dollars
- **Receiver:** (string) username of the user who received the order
- **Rating:** (int) how the user rated their order on a scale of 1-10 (10 being most satisfied)
- **With Friends:** (boolean) whether or not the user placed the order with friends

## heights

Here is a preview of the table `heights`:

- `height` is the **float** height of an individual, measured in inches
- `enrollee` is a **string** containing either “high school” or “college”

| height | enrollee    |
|--------|-------------|
| 68.5   | college     |
| 62.1   | high school |
| 66.5   | college     |

## rentals

Here is a preview of the table `rentals`:

- `date` contains the **string** date on which the rental data were collected
- `weekend` is a **boolean** corresponding to whether the day fell on a weekend
- `num_kayaks` is the **integer** number of kayak rentals on that particular day
- `num_boards` is the **integer** number of windsurfing board rentals on that particular day
- `wind` is the **integer** wind speed on that particular day, in miles per hour

| date  | weekend | num_kayaks | num_boards | wind |
|-------|---------|------------|------------|------|
| 04/29 | True    | 35         | 19         | 4    |
| 05/01 | True    | 26         | 24         | 22   |
| 05/02 | False   | 17         | 14         | 10   |

## skiers

Here is a preview of the table **skiers**:

| Name    | Sport     | Height (in) | Downhill Time (s) | Favorite Resort |
|---------|-----------|-------------|-------------------|-----------------|
| James   | Ski       | 71          | 90.52             | Vail            |
| Eunice  | Ski       | 66          | 93.64             | Beaver Creek    |
| Oscar   | Snowboard | 69          | 89.77             | Heavenly        |
| Rebecca | Snowboard | 68          | 91.01             | Palisades       |
| Ciara   | Ski       | 70          | 101.34            | Park City       |

... (40 rows omitted)

## restaurant

Here is a preview of the table **restaurant**:

| Restaurant          | Type   | Distance from Campus |
|---------------------|--------|----------------------|
| Round Table         | Pizza  | 2.2                  |
| Panera              | Bagels | 2.3                  |
| Feng Cha            | Boba   | 0.13                 |
| Boba Guys           | Boba   | 1.5                  |
| Berkeley Thai House | Thai   | 0.15                 |

... (306 rows omitted)

## transport

Here is a preview of the table **transport**:

| Restaurant | Transportation | Time |
|------------|----------------|------|
| Panera     | Bus            | 27   |
| La Burrita | Walk           | 5    |
| Panera     | Walk           | 62   |
| Boba Guys  | Drive          | 10   |
| Panera     | Drive          | 12   |

... (1492 rows omitted)

## bananas

Here is a preview of the table **bananas**:

| Date     | Customers | Sales | Session | Topping    |
|----------|-----------|-------|---------|------------|
| 04/16/20 | 46        | 243   | School  | Fudge      |
| 06/15/20 | 54        | 233   | Summer  | Nuts       |
| 09/01/20 | 65        | 254   | School  | Pink Candy |
| 10/05/20 | 48        | 278   | School  | Nuts       |

... (112 rows omitted)

Each row corresponds to a day in 2020 in which the frozen banana stand was open. The table contains the following columns:

- *Date*: (string) the day the stand was open
- *Customers*: (int) the number of total paying customers received that day
- *Sales*: (int) the number of total bananas sold that day
- *Session*: (string) 'School' if high school is in session on that day, 'Summer' if it's not
- *Topping*: (string) the most popular topping sold that day